Farzad Mehri

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Ph.D. Student | Network Optimization, Satellite Networks, Spectrum Coexistence, Space Data Center

Education

NC State University — Raleigh, NC
Ph.D. in Computer Engineering, GPA: 4.0/4.0 — *Expected 05/2028*

University of Tehran — Tehran, Iran
M.Sc. in Electrical Engineering, GPA: 3.76/4.0 — *09/2022*

University of Tehran — Tehran, Iran
B.Sc. in Electrical Engineering — *09/2019*

Publications

- **Farzad Mehri**, Dara Ron, Satyaki Roy, Nishanth Sastry, and Vijay K. Shah. "SkyShare: Constellation-wide Sky Sharing for LEO–Radio Astronomy Coexistence." **ACM MobiCom 2026**.
- **Farzad Mehri**, et al. "Proactive Xn-Route-Aware Conditional Handover Planning for Regenerative LEO Networks." **Under review**.
- Mohsen Shekarisaz, Fatemeh Talebian, Marjan Jabariani, **Farzad Mehri**, Fathiyeh Faghih, and Mehdi Kargahi. "Program Energy-Hotspot Detection and Removal: A Static Analysis Approach." **IEEE RTEST 2020**.

Research Experience

NextG Lab, NC State University — Raleigh, NC
Research Assistant — *08/2024 – Present*

- Developed **SkyShare**, a constellation-wide LEO spectrum-sharing system; formulated beam scheduling as a mixed-integer linear program and designed **SkySched**, a scalable min-cost-flow scheduler that achieved over 99% of the optimal objective and reduced unserved cells by 90.7% on average across 25 worldwide radio-astronomy sites.
- Developed **ProCHO**, a proactive mobility planner for regenerative LEO networks; combined time-layered path optimization, visible-satellite clustering, Lagrangian relaxation, and per-UE shortest paths to reduce handover interruption by 18.8% versus the strongest baseline at comparable downlink rate.
- Built Python-based system and Monte Carlo simulators for TLE-derived constellations, UE–satellite geometry, time-varying routing, 3GPP-aligned propagation, link budgets, SINR/MCS, outage, handover, and capacity-constrained scheduling; demonstrated near-linear planning scalability to 8,000 UEs.
- Extended **SimAI** with a C++ satellite network backend and an orbital constellation model (Walker/J2, SGP4/TLE, time-varying +GRID links) to simulate torus, hierarchical, and ring all-reduce performance in space LLM training.
- Evaluated inter-satellite topology, shortest-path latency, bandwidth, and link churn using NS-3, Mininet, and SpaceNet; building an O-RAN/5G NTN testbed with OpenAirInterface, FlexRIC, RFSim, and USRP B210 SDRs.

ECE Department, NC State University — Raleigh, NC
Teaching Assistant, Computer Microarchitecture — *08/2024 – 12/2024*

- Supported a 200+ student course and implemented C++ simulators for superscalar out-of-order execution, caches, and branch prediction to study architecture-level bottlenecks and performance tradeoffs.

Dependable/Distributed Real-Time Systems Lab — Tehran, Iran
Research Assistant — *08/2021 – 08/2022*

- Designed a LoRaWAN-based MAC protocol for soft real-time communication, combining compact gateway-issued schedules with slotted ALOHA to support periodic and sporadic traffic under limited downlink capacity and imperfect clock synchronization.
- Developed scheduling heuristics and evaluated throughput under varying traffic loads and synchronization accuracy; simulations showed sustained throughput under overload, avoiding the throughput collapse observed with pure ALOHA.
- Collected and analyzed Raspberry Pi power-spectrum traces for static detection and removal of program energy hotspots.

Technical Skills

Algorithms / Optimization: Convex optimization, time-expanded MILP scheduling, min-cost flow, shortest paths, Lagrangian relaxation, distributed and real-time systems, performance modeling, Monte Carlo simulations
Programming / Research Software: Python, C, C++, MATLAB, Linux, Git/GitHub, Pytest, CI/CD, NS-3, Mininet
Satellite / Networking: LEO constellations, orbital prediction/TLEs, inter-satellite links, routing, 5G NR/LTE, 3GPP NTN, O-RAN, handover, spectrum sharing, link budgets, SDR
AI / Machine Learning: Backpropagation, neural-network fundamentals, classical machine-learning algorithms, GNNs, reinforcement learning, random forests, ONNX, LLM architectures and training concepts, all-reduce collectives, CNNs

Relevant Coursework

Mathematics / Optimization: Convex Optimization; Advanced Computer Mathematics; Stochastic Processes; Statistical Inference; Introduction to Intelligent Systems
Distributed / Networked Systems: Fundamentals of Blockchain and Distributed Computing; Internet Protocols; Advanced Internet Protocols; Advanced Computer Networking; Advanced Next-Generation Network Design
Computer / Wireless Systems: Computer Microarchitecture; Performance and Security of Advanced Microarchitecture; LTE and 5G Communications; Open RAN: Foundations and Applications; Antenna Theory and Laboratory

Additional Engineering Experience

Holoo Software Tehran, Iran
IoT Engineer *12/2022 – 08/2023*

- Developed embedded C firmware for Wi-Fi, Bluetooth, and Zigbee devices and Python/Django middleware for device discovery, command/control, telemetry, and mobile-app integration over MQTT/REST.
- Automated API and integration testing with Pytest and CI/CD; deployed and validated the system on Raspberry Pi hardware and custom PCBs.

Datis Tehran, Iran
Embedded Developer *03/2022 – 08/2023*

- Simulated an MPPT control algorithm in C++, implemented it in C on an STM32 industrial drive, and validated it through hardware-in-the-loop measurements and physical load testing.
- Implemented low-overhead flash storage and real-time, DMA-driven audio playback; debugged an OTA firmware-update path spanning BLE, Ethernet, and a mobile application.

Selected Technical Projects

- **Computer architecture and security:** Implemented an entry-level RISC processor in Verilog; developed C++ microbenchmarks to reverse-engineer the cache hierarchy and other microarchitectural characteristics of an Intel CPU; and evaluated Intel Advanced Matrix Extensions (AMX) under speculative execution for microarchitectural attack analysis.
- **Machine learning and optimization:** Built a graph neural network for optimal-power-flow prediction and integrated random-forest models through ONNX into mobility-control experiments.
- **Deep learning for medical imaging (B.Sc. thesis):** Fine-tuned RetinaNet, YOLO, and Mask R-CNN using transfer learning and online data augmentation for mammographic mass detection, segmentation, and benign/malignant classification on CBIS-DDSM; evaluated localization and sensitivity–false-positive tradeoffs.
- **Embedded AI and vision:** Ported the comma.ai driving stack to an NVIDIA Jetson Nano by adapting ARM-specific accelerated operations for CUDA execution and integrating a custom camera using NVIDIA vision-processing libraries, enabling end-to-end operation on resource-constrained hardware.

Selected Honors

- **1st Place Winner**, OpenAirInterface (OAI) Foundation Hands-on Workshop Hackathon (2026). Recognized for engineering a 5G NR MAC slicing policy in OAI repository.
- NC State University College of Engineering Graduate Merit Award.
- Ranked in top 0.7% among approximately 167,000 examinees in Iran's national university entrance examination, 2015.